

Data Link Layer

DLL: Responsible for error-free transfer of data frames between adjacent nodes.

Framing

Dividing stream of bits into manageable data units called frames.

Flow Control

Restricting amount of data sender can send before waiting for acknowledgment.

Error Control

Detecting and retransmitting damaged or lost frames (ARQ).

Stop-and-Wait ARQ:

Sender sends one frame, waits for ACK before sending the next.

Go-Back-N ARQ:

Sender sends multiple frames. If one is lost, retransmits that frame and all subsequent frames.

Selective Repeat ARQ:

Only the damaged/lost frame is retransmitted.

MAC Sublayer:

- CSMA/CD: Used in wired LANs (Ethernet) to detect collisions.
- CSMA/CA: Used in wireless LANs (Wi-Fi) to avoid collisions.